Classification

Model evaluation

Quick modeling of high_revenue on customers_labeled / Models / Random forest

Report | Predicted data | Charts | PUBLISH | DEPLOY | ACTIONS

Summary

INTERPRETATION

- Decision trees
- Variables importance
- Partial dependence
- Subpopulation analysis
- Individual explanations
- Interactive scoring

PERFORMANCE

- Confusion matrix
- Decision chart
- Lift charts
- Calibration curve
- ROC curve
- Density chart
- Metrics and assertions

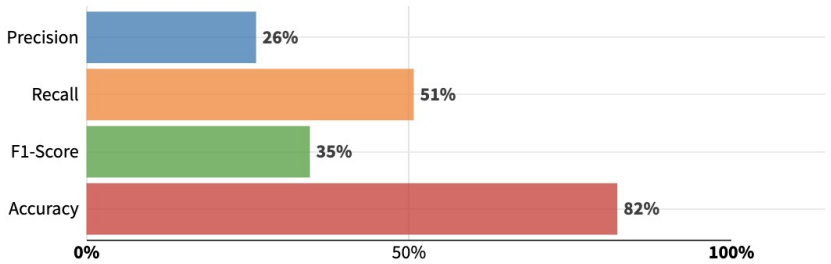
MODEL INFORMATION

- Data preparation
- Features
- Algorithm
- Hyperparameter optimization
- Training information

Threshold (cut-off) 0 — 1 `0.600` BACK TO OPTIMAL*

Display: Record count

	Predicted `True`	Predicted `False`	Total
Actually `True`	102	99	201
Actually `False`	286	1690	1976
Total	388	1789	2177



Cost matrix

If model predicts `True`	and value is `True`	the gain is	1	×	102	=	102.00
	but value is `False`	the gain is	-0,3	×	286	=	-85.80
Model predicts `False`	and value is `False`	the gain is	0	×	1690	=	0.00
	but value is `True`	the gain is	0	×	99	=	0.00
		Average gain per record	**0.01**	×	2177	=	16.20

* "Optimal" cut was found by optimizing for **F1 Score**.

Description

A classifier produces a probability that a given object belongs to the class (e.g. that `high_revenue` is `True`). The threshold (or "cut-off") is the number beyond which the prediction is considered positive. If set too low, it may predict `True` too often, if set too high, too rarely.

One way to assess a classification model's performance is to use a "confusion matrix", which compares actual values (from the test set) to predicted values. Be careful though, the figures are highly dependent on the probability cutoff chosen to classify a record. Depending on your use case, you might want to adjust the cutoff to optimize a specific metric.

Metrics definitions

Precision: Proportion of correct ("positive") predictions among "positive" predictions.
Recall: Proportion of (correct) "positive" predictions among "positive" records.
F1-score: Harmonic mean between precision and recall.
Accuracy: Proportion of correct predictions among all records ("positive" and "negative"). It is less informative than F1-score for unbalanced datasets.

Cost matrix gain

You can also evaluate the average gain per record that the test set would yield by specifying a gain for each outcome, e.g. you win $1 for each correct prediction of `True` but $-0.4 (i.e. you lose $0.4) if that prediction turns out to be incorrect.

Medical diagnosis

Healthy

Sick

Confusion matrix

Caution: positive/negative like in e.g. AIDS:

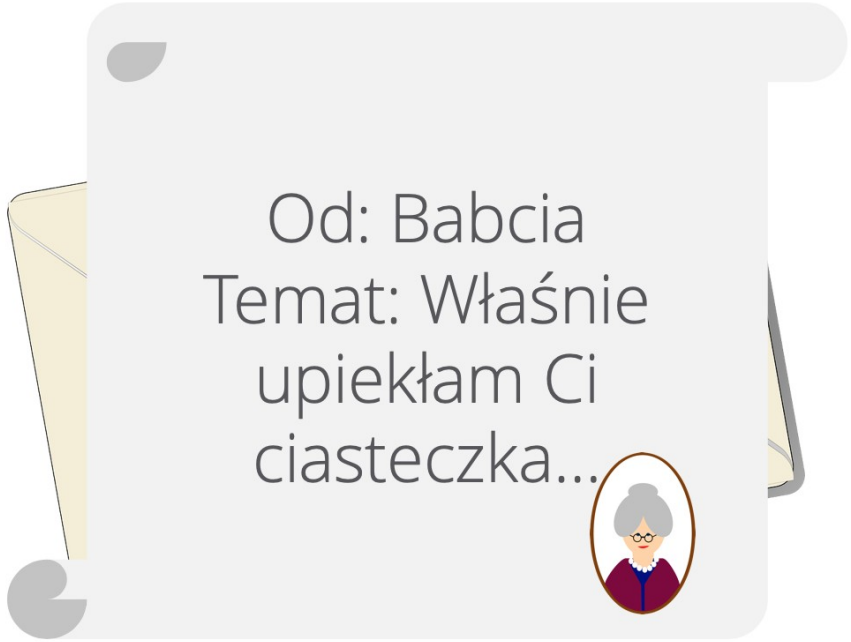
Positive: is sick
Negative: is not sick

	Diagnosis (model): sick	Diagnosis (model): healthy
Reality: sick	True Positive	False Negative
Reality: healthy	False Positive	True Negative

For 10 000 patients

	Diagnosis (model): sick	Diagnosis (model): healthy
Reality: sick	1000 True Positive	200 False Negative
Reality: healthy	800 False Positive	8000 True Negative

SPAM



Nie SPAM

SPAM

Confusion matrix

Antispam filter = SPAM classifier

	Sent to SPAM	Left in the inbox
SPAM	True Positive	False Negative
Not SPAM	False Positive	True Negative

Diagnosis of 10 000 e-mail's

	Sent to SPAM	Left in the inbox
SPAM	1000 True Positive	200 False Negative
Not SPAM	800 False Positive	8000 True Negative

Accuracy

Accuracy

Of all patients, how many percent did we diagnose correctly?

$$Accuracy = \frac{1{,}000 + 8{,}000}{10{,}000} = 90\%$$

	Diagnosis (model): sick	Diagnosis (model): healthy
Reality: sick	1000	200
Reality: healthy	800	8000

```
from sklearn.metrics import accuracy_score
accuracy_score (y_true, y_pred)
```

```
In [1]: from sklearn.metrics import accuracy_score

In [2]: y_true = [0, 0, 0, 0, 1, 1, 1, 1]
        y_pred = [0, 0, 1, 0, 1, 0, 1, 1]

In [3]: accuracy_score(y_true, y_pred)

Out[3]:
```

sklearn.metrics.accuracy_score

`sklearn.metrics.` **`accuracy_score`** (*y_true, y_pred, normalize=True, sample_weight=None*) [source]

Accuracy classification score.

In multilabel classification, this function computes subset accuracy: the set of labels predicted for a sample must *exactly* match the corresponding set of labels in y_true.

Read more in the User Guide.

Parameters:

y_true : ***1d array-like, or label indicator array / sparse matrix***

Ground truth (correct) labels.

y_pred : ***1d array-like, or label indicator array / sparse matrix***

Predicted labels, as returned by a classifier.

normalize : ***bool, optional (default=True)***

If `False`, return the number of correctly classified samples. Otherwise, return the fraction of correctly classified samples.

sample_weight : ***array-like of shape = [n_samples], optional***

Sample weights.

Returns:

score : ***float***

If `normalize == True`, return the fraction of correctly classified samples (float), else returns the number of correctly classified samples (int).

The best performance is 1 with `normalize == True` and the number of samples with `normalize == False`.

See also:

`jaccard_score`, `hamming_loss`, `zero_one_loss`

Notes

In binary and multiclass classification, this function is equal to the `jaccard_score` function.

Examples

```
>>> from sklearn.metrics import accuracy_score
>>> y_pred = [0, 2, 1, 3]
>>> y_true = [0, 1, 2, 3]
>>> accuracy_score(y_true, y_pred)
0.5
>>> accuracy_score(y_true, y_pred, normalize=False)
2
```

In the multilabel case with binary label indicators:

```
>>> import numpy as np
>>> accuracy_score(np.array([[0, 1], [1, 1]]), np.ones((2, 2)))
0.5
```

In [1]:
```
from sklearn.metrics import accuracy_score
```

In [2]:
```
y_true = [0, 0, 0, 0, 1, 1, 1, 1]
y_pred = [0, 0, 1, 0, 1, 0, 1, 1]
```

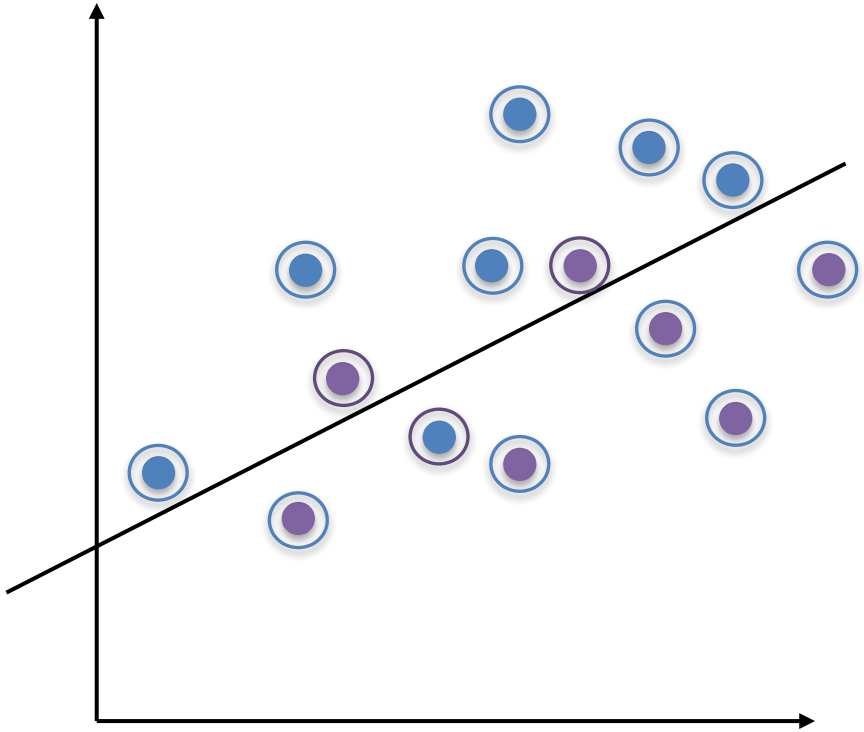
In [3]:
```
accuracy_score(y_true, y_pred)
```

Out[3]:
```
0.75
```

Accuracy

Of all the points, how many did we classify correctly?

$$Accuracy = \frac{Correctly\ classified\ points}{All\ the\ points}$$



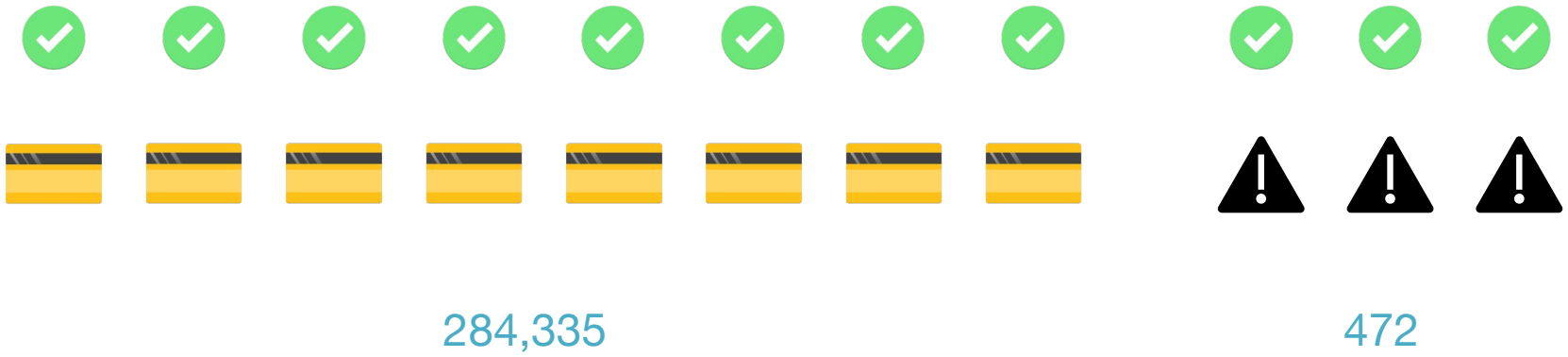
$$= \frac{11}{11 + 3}$$

$$= \frac{11}{14}$$

$$= 78.57\%$$

When is accuracy a wrong measure?

Credit Card Frauds

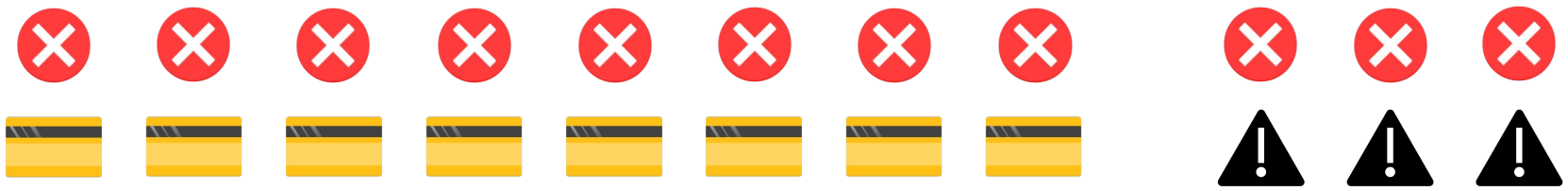


MODEL: All transactions are OK

$$\text{Accuracy} = \frac{284{,}335}{284{,}887} = 99.83\%$$

PROBLEM: We didn't identify any fraud!

Credit Card Frauds



284,335 472

MODEL: All transactions are frauds

Great! Now I've caught all the scams....

PROBLEM: I classified all valid transactions as fraudulent....

The solution: recall and precision...

Medical diagnosis

Which is worse (which error we want to minimize):

False Negative

or

False Positive

	Diagnosis (model): sick	Diagnosis (model): healthy
Reality: sick	True Positive	False Negative
Reality: healthy	False Positive	True Negative

SPAM filter

Which is worse (which error we want to minimize):
False Negative
or
False Positive

	Sent to SPAM	Left in inbox
SPAM	True Positive	False Negative
Not SPAM	False Positive	True Negative

Medical diagnosis

FALSE POSITIVES OK

FALSE NEGATIVES NOT OK

Goal: find everyone who is sick

OK, if not everyone we diagnose as sick is actually healthy

High recall

SPAM filter

FALSE POSITIVES NOT OK

FALSE NEGATIVES OK

It is important that you do not put important information into SPAM

You don't have to put all your SPAM in SPAM

High precision

Recall

True Positive Rate

Diagnosis of 10 000 patients

Rrecall:

Of all the actual sick patients, how many were diagnosed with the disease?

$$\text{Recall} = \frac{1{,}000}{1{,}000 + 200}$$

$$= 83{,}3\%$$

	Diagnosis (model): sick	Diagnosis (model): healthy
Reality: sick	1 000	200 ✕
Reality: healthy	800	8 000

Antispam filter

Recall:

Of all the emails that were actually SPAM, how many did we send to SPAM?

$$\text{Recall} = \frac{100}{100 + 170}$$

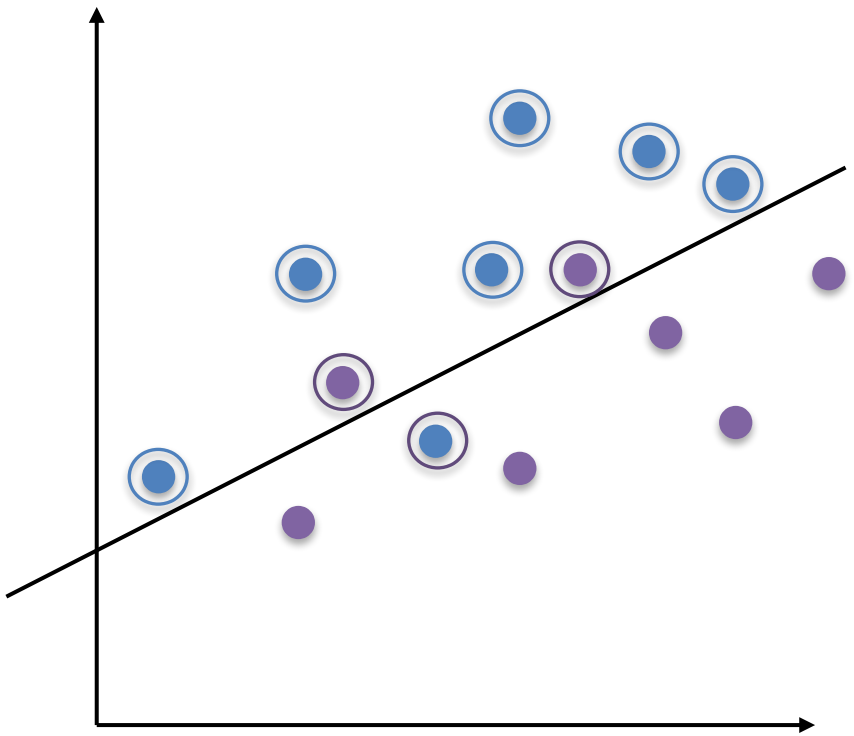
$$= 37.0\%$$

	Sent to SPAM	Left in inbox
SPAM	100	170
Not SPAM	30 ❌	700

Recall

Of all the points that were actually true, how many did we rate as true?

$$Recall = \frac{Points\ correctly\ classified\ by\ us\ as\ true}{All\ points\ which\ are\ true}$$



$$= \frac{6}{7}$$

$$= 85{,}7\%$$

Precision

Positive Predictive Value

Diagnosis of 10 000 patients

Precision:

Of all the patients we diagnosed with the disease, how many were actually sick?

$$\text{precision} = \frac{1{,}000}{1{,}000 + 800}$$

$$= 55.6\%$$

	Diagnosis (model): sick	Diagnosis (model): healthy
Reality: sick	1 000	200 ✕
Reality: healthy	800	8 000

Antispam filter

Precision:

Of all the emails we sent to SPAM, how many were actually SPAM?

$$\text{precision} = \frac{100}{100 + 30}$$

$$= 76{,}9\%$$

	Wysłany do SPAMu	Pozostawiony w skrzynce
SPAM	100	170
Nie SPAM	30 ❌	700

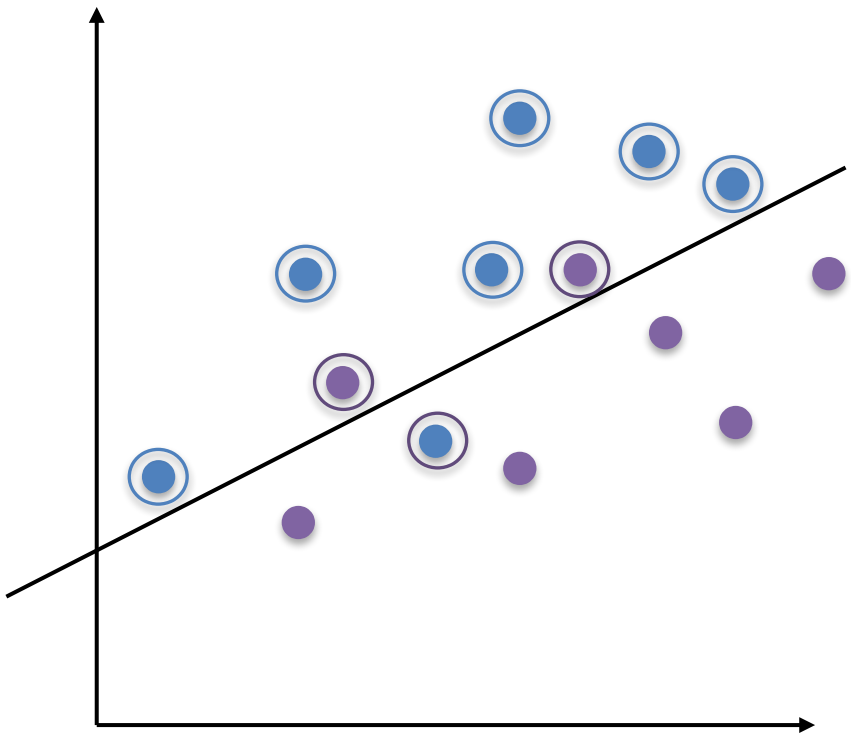
Precision

Of all the points we rated as true, how many were actually true?

$$Precision = \frac{Actually\ true\ points}{Points\ classified\ as\ true}$$

$$= \frac{6}{8}$$

$$= \cdots \%$$



	Wysłany do SPAMu	Pozostawiony w skrzynce
SPAM	100	170
Nie SPAM	30 ✕	700

precision

true positives

/

all predicted as positive

	Diagnosis (model): sick	Diagnosis (model): healthy
Reality: sick	1 000	200 ✕
Reality: healthy	800	8 000

recall

true positives

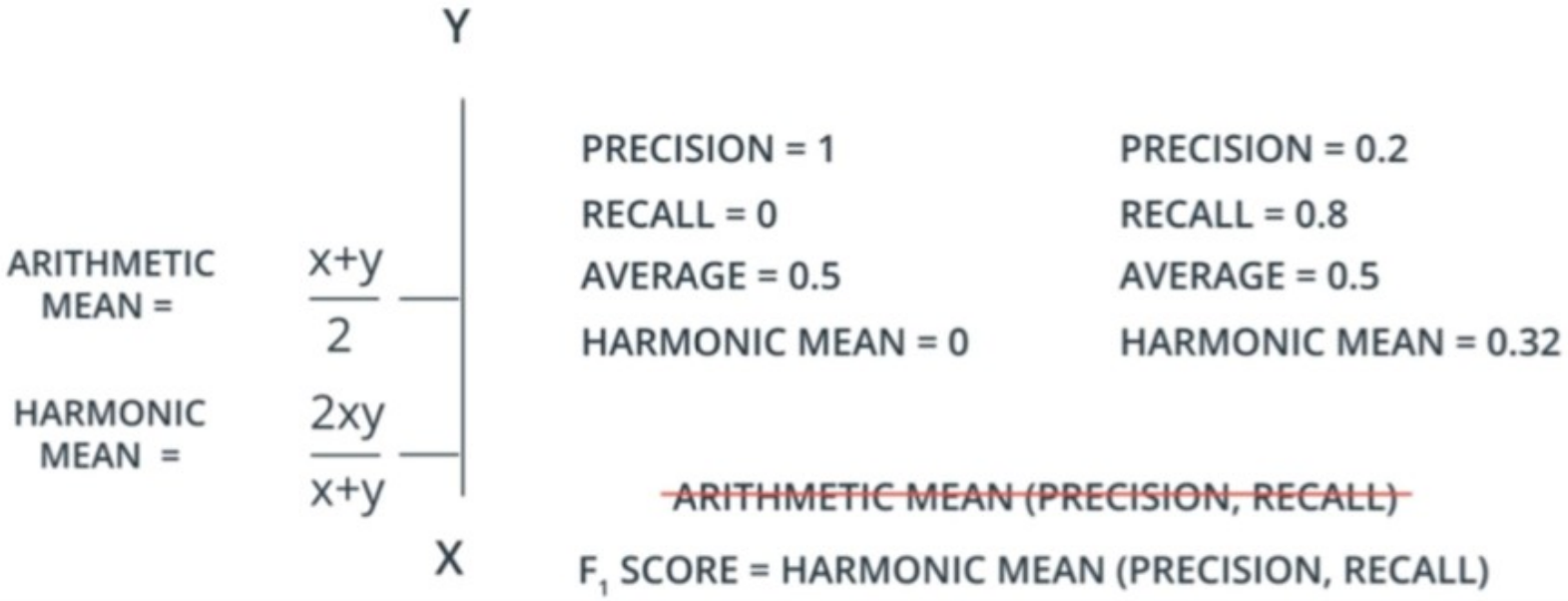
/

all predicted as positive

F1: synthesis of precision and recall

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

Harmonic mean



F1 in medical diagnosis

Precision = 55,6%

Recall = 83,3%

F1 = ?

F1 = 66,67%

F_beta: F1 generalization

The choice of beta depends on the problem....

$$F_1 \text{ SCORE} = 2 \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$F_\beta = (1 + \beta^2) \cdot \frac{\text{precision} \cdot \text{recall}}{(\beta^2 \cdot \text{precision}) + \text{recall}}$$

PRECISION $F_{0.5}$ SCORE F_1 SCORE F_2 SCORE RECALL

An algorithm for detecting faulty components in a spacecraft?

What factor would you choose to evaluate...

β = ???

β = 0,5

β = 1

✓ **β = 2**

What factor would you choose to evaluate...

β = ???

A system that sends SMS text messages with recommendations of movies the customer might like?

β = 0,5

✓ β = 1

β = 2

A system for sending advertising e-mails

What factor would you choose to evaluate...

β = ???

β = 0,5

✓ **β = 1**

β = 2

ROC

Receiver Operating Characteristics

Medical diagnosis

Healthy

Sick

Medical diagnosis

Healthy: no nodules present

Sick: nodules present

Recall

True positive rate

Of all the actual sick patients, how many were diagnosed with the disease?

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Recall} = \frac{1{,}000}{1{,}000 + 200} = 83{,}3\%$$

	Diagnosis (model): sick	Diagnosis (model): healthy
Reality: sick	1 000	200 ✕
Reality: healthy	800	8 000

Precision

Positve predicted value

Of all the patients we diagnosed with the disease, how many were actually sick?

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Precision} = \frac{1{,}000}{1{,}000 + 800} = 55{,}6\%$$

	Diagnosis (model): sick	Diagnosis (model): healthy
Reality: sick	1 000	200 ✕
Reality: healthy	800	8 000

Our model returns probabilities

How to translate them into diagnosis?

Probability = 0.6

?

Healthy
no nodules present

Sick
nodules present

How to choose probability threshold?

healthy < 0.5 < sick?
healthy < 0.6 < sick?
healthy < 0.7 < sick?

Probability = ???

?

Healthy
no nodules present

Sick
nodules present

Demo

We have 10 CTR images

For each we collect:

- **Model prediction (= probability)**
- **Label (ground truth)**

Next, we change the threshold:

10, 20, 30, …, 90

and for each such a threshold compute:
Recall

Precision

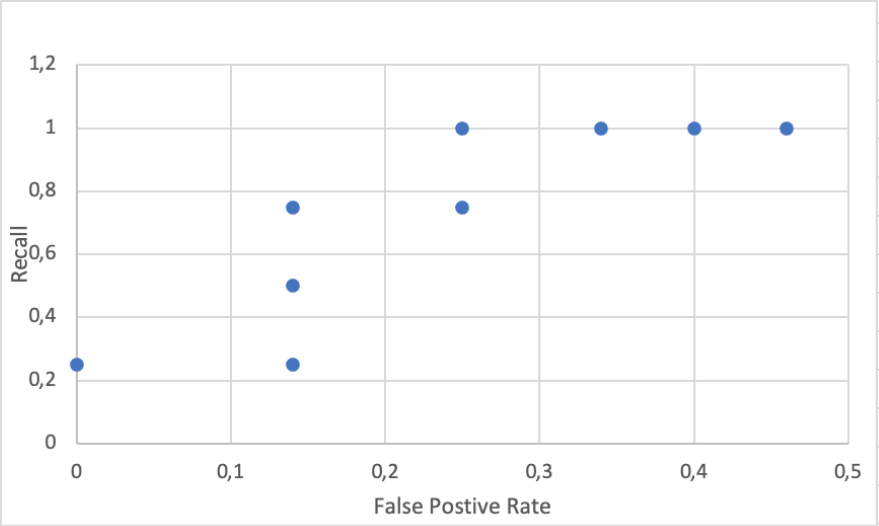
$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

prob	label	prediction	TP	FP	FN
9	0	0	0	0	0
19	0	0	0	0	0
29	0	0	0	0	0
39	0	0	0	0	0
49	1	0	0	0	1
59	0	0	0	0	0
69	1	0	0	0	1
79	1	0	0	0	1
89	0	0	0	0	0
99	1	1	1	0	0
True Negative >	6		**1**	**0**	**3**

threshold	90	
Precision	1	= TP/(TP + FP)
Recall (true positive rate)	0,25	= TP/(TP + FN)
False-positive rate	0	= FP/(FP + TN)

threshold	**false p**	**recall**	**precision**
10	0,46	**1**	0,44
20	0,4	1	0,50
30	0,34	1	0,57
40	0,25	1	0,67
50	0,25	0,75	0,60
60	0,14	0,75	0,75
70	0,14	0,5	0,67
80	0,14	0,25	0,50
90	0	0,25	1,00



ROC

Receiver Operating Characteristics

Brak choroby

Choroba

ROC AUC

AUC represents the probability that a random positive (green) example is positioned below a random negative (red) example.

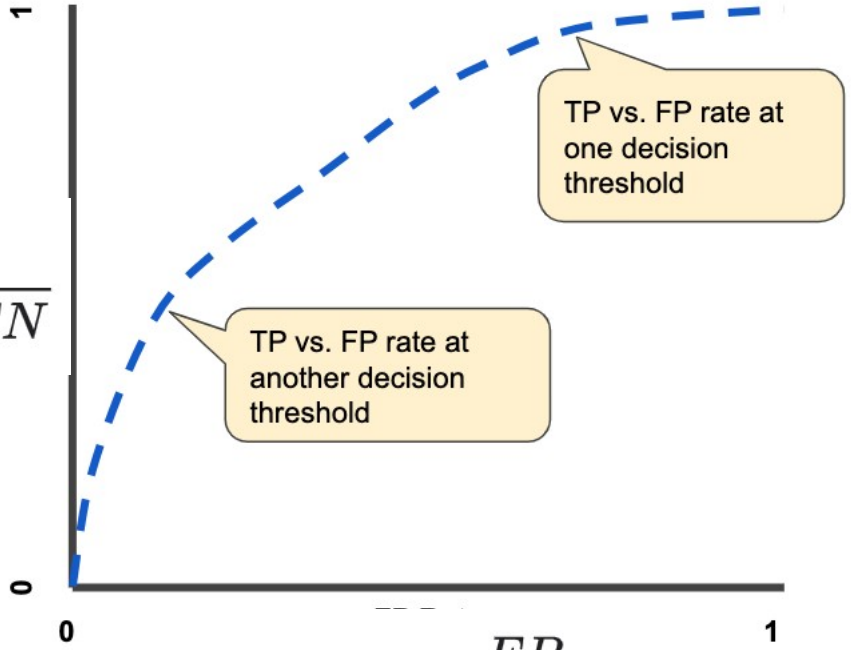
AUC ranges in value from 0 to 1. A model whose predictions are 100% wrong has an AUC of 0.0; one whose predictions are 100% correct has an AUC of 1.0.

Benefits:

- AUC is **scale-invariant**. It measures how well predictions are ranked, rather than their absolute values.
- AUC is **classification-threshold-invariant**. It measures the quality of the model's predictions irrespective of what classification threshold is chosen.

prawda	wynik klasyfikacji
0	P = 0.0
0	
0	
0	
1	
0	
1	
1	
0	
1	P = 1.0

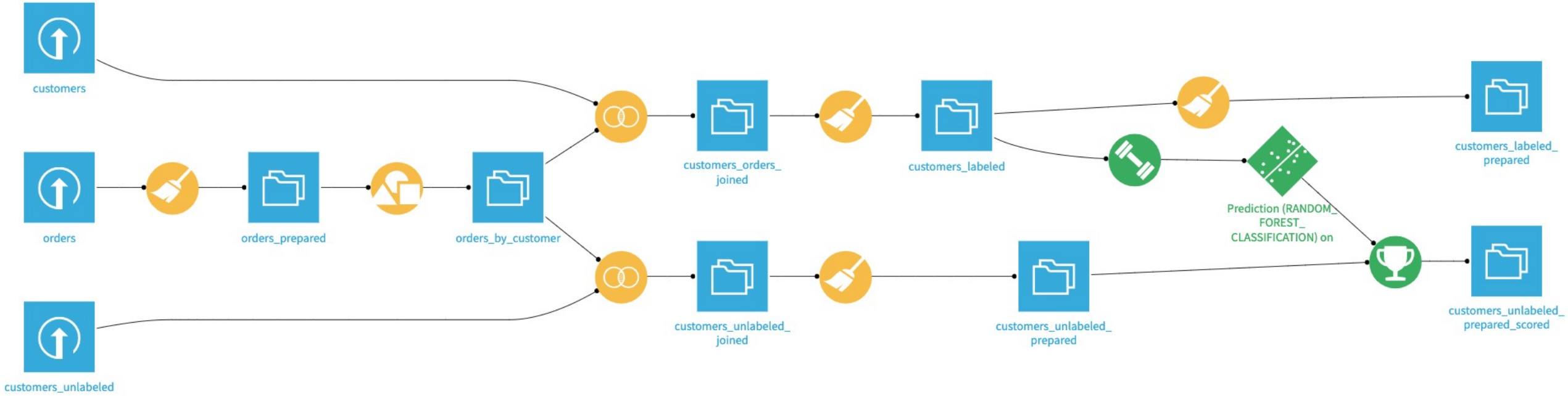
$$TPR = \frac{TP}{TP + FN}$$



$$FPR = \frac{FP}{FP + TN}$$

Goal

Based on customer behavior in an e-commerce store, create a model that correctly assesses the likelihood that a new customer will be high-margin



Prediction outcome

3 additional columns:

Probability False

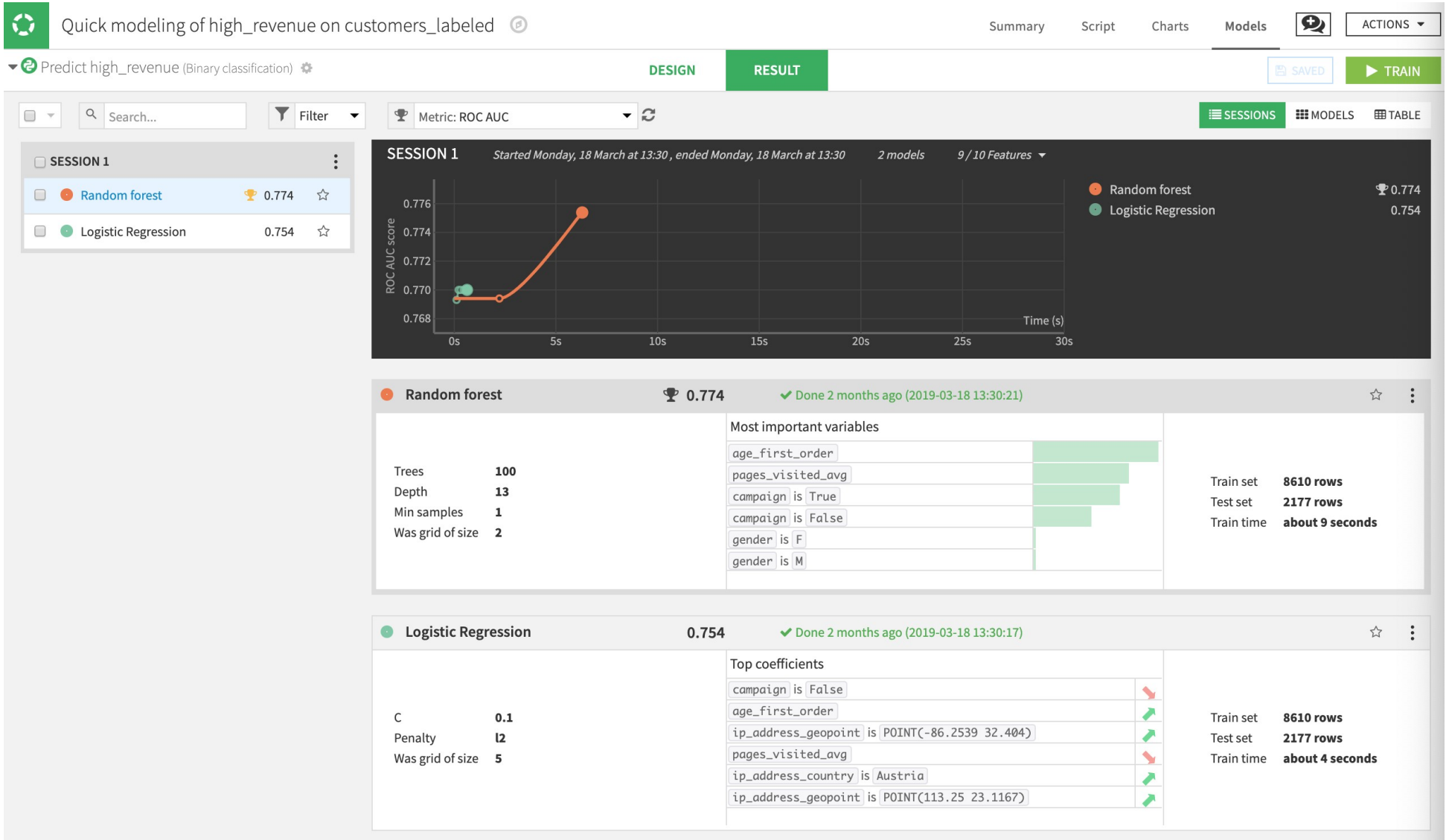
Probability True

Prediction: True or False

On what basis did the algorithm made a prediction???

user_agent_brand	user_agent_os	ip_address_country	ip_address_geopoint	campaign	pages_visited_avg	proba_False	proba_True	prediction
string	string	string	string	boolean	double	double	double	string
Text	Text	Country	GeoPoint	Boolean	Decimal	Decimal	Decimal	Boolean
Chrome	Windows	United States	POINT(-75.6203 39.9239)	False	14.0	0.7026505317961358	0.29734946820386426	False
Chrome	Windows			True	12.0	0.3911949244042885	0.6088050755957115	True
Chrome	Windows	Japan	POINT(139.7677 35.6427)	True	13.0	0.6592170051793084	0.34078299482069163	False
Chrome	Windows	China	POINT(110.3417 20.0458)	False	14.0	0.8186948094602139	0.1813051905397861	False
Chrome	MacOS X	Finland	POINT(24.9375 60.1708)	False	12.0	0.6467799892209887	0.35322001077901133	False
Chrome	MacOS X	United States	POINT(-78.1704 38.7163)	False	15.0	0.7719956518968101	0.2280043481031899	False
IE	Windows	United States	POINT(-117.1649 32.8072)	False	20.0	0.7316943707097567	0.2683056292902432	False
Chrome	Windows	Australia	POINT(143.2104 -33.494)	True	12.0	0.5165851713705042	0.4834148286294958	False
Chrome	Windows	United States	POINT(-97.822 37.751)	False	12.0	0.632670373159543	0.367329626840457	False
Safari	MacOS X	Republic of Korea	POINT(128.6811 35.2281)	False	13.0	0.7411251165166426	0.25887488348335747	False
Safari	MacOS X	United States	POINT(-122.2568 37.8672)	False	9.0	0.5799103645434219	0.42008963545657807	False
Firefox	Windows	United States	POINT(-110.8313 41.2355)	True	12.0	0.5323197828556991	0.46768021714430086	False
Chrome	MacOS X	United States	POINT(-80.1711 26.1882)	False	18.0	0.9064282261291514	0.09357177387084857	False
Firefox	Windows	Australia	POINT(151.1982 -33.8612)	False	19.0	0.7580311944007768	0.24196880559922318	False
Edge	Windows	Netherlands	POINT(5.1191 52.0938)	True	21.0	0.7401145238617263	0.25988547613827373	False
Edge	Windows	Spain	POINT(-8.396 43.3713)	False	10.0	0.845483377375744	0.15451662262425597	False
Chrome	Windows	France	POINT(1.535 45.158)	False	10.0	0.6167737249805421	0.38322627501945794	False
Chrome	MacOS X	United States	POINT(-97.822 37.751)	True	14.0	0.878866152203821	0.12113384779617892	False
Chrome	Windows	United States	POINT(-78.6253 35.7977)	False	13.0	0.7481391446036973	0.25186085539630265	False
Safari	MacOS X	United States	POINT(-74.1741 40.7355)	True	14.0	0.6264792813854463	0.37352071861455366	False
Chrome	Windows	Germany	POINT(9.491 51.2993)	False	8.0	0.622427656685175	0.3775723433148251	False

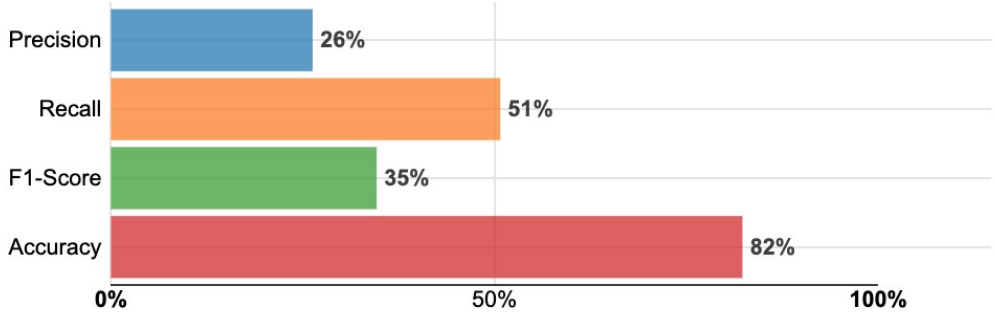
Model evaluation



Threshold (cut-off) 0 — 1 **0.600** BACK TO OPTIMAL*

Display: Record count

	Predicted `True`	Predicted `False`	Total
Actually `True`	102	99	201
Actually `False`	286	1690	1976
Total	388	1789	2177



Cost matrix

If model predicts `True`	and value is `True`	the gain is	1	×	102	=	102.00
	but value is `False`	the gain is	-0,3	×	286	=	-85.80
Model predicts `False`	and value is `False`	the gain is	0	×	1690	=	0.00
	but value is `True`	the gain is	0	×	99	=	0.00
		Average gain per record	**0.01**	×	2177	=	16.20

* "Optimal" cut was found by optimizing for **F1 Score**.

Description

A classifier produces a probability that a given object belongs to the class (e.g. that `high_revenue` is `True`). The threshold (or "cut-off") is the number beyond which the prediction is considered positive. If set too low, it may predict `True` too often, if set too high, too rarely.

One way to assess a classification model's performance is to use a "confusion matrix", which compares actual values (from the test set) to predicted values. Be careful though, the figures are highly dependent on the probability cutoff chosen to classify a record. Depending on your use case, you might want to adjust the cutoff to optimize a specific metric.

Metrics definitions

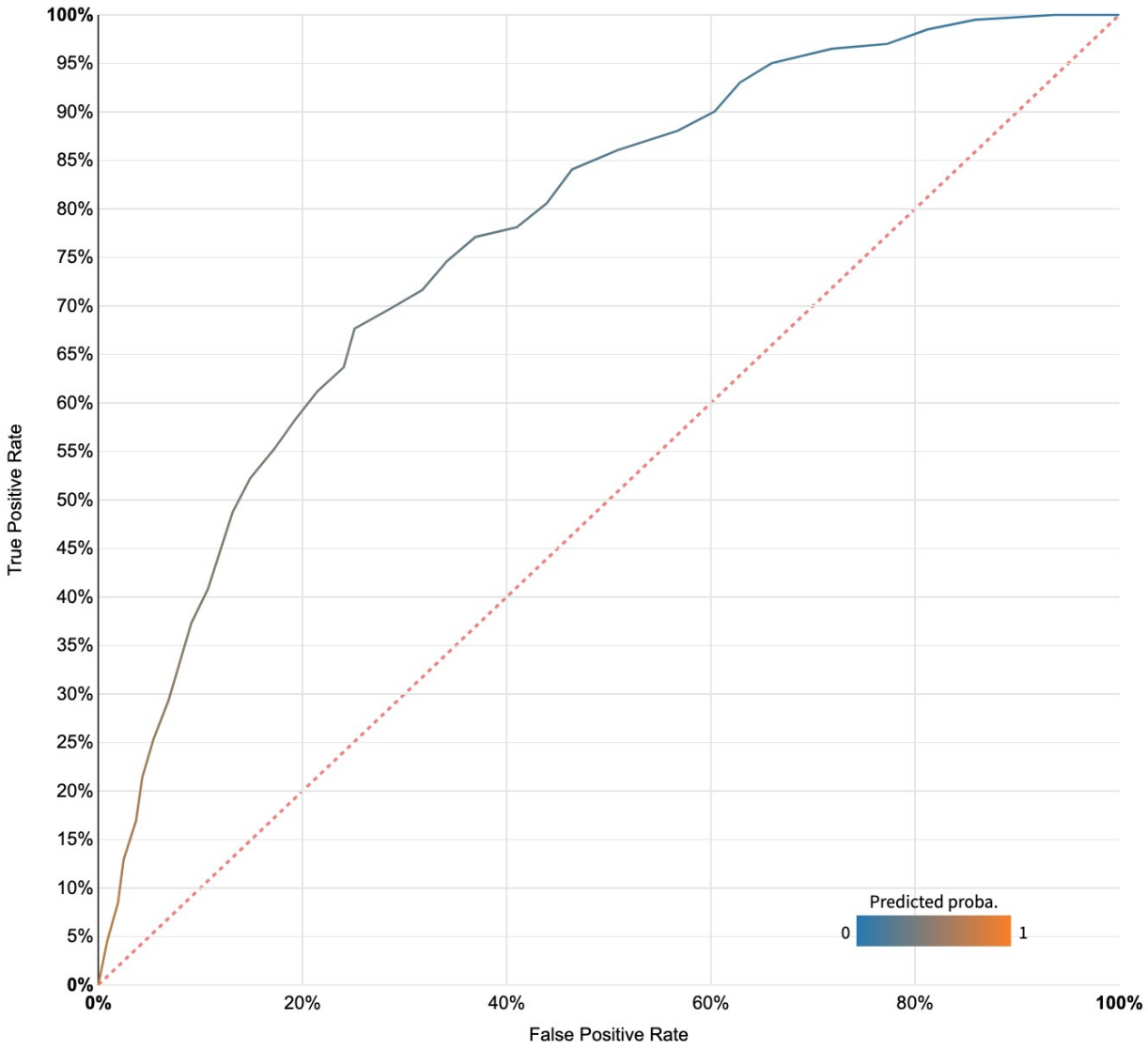
Accuracy: Proportion of correct predictions (positive and negative) in the sample.
Precision: Proportion of correct "positive" predictions in the sample.
Recall: Proportion of "positive" actual records correctly predicted as "positive".
F1-score: Harmonic mean between precision and recall. More informative than Accuracy for unbalanced datasets.

Cost matrix gain

You can also evaluate the average gain per record that the test set would yield by specifying a gain for each outcome, e.g. you win $1 for each correct prediction of `True` but $-0.4 (i.e. you lose $0.4) if that prediction turns out to be incorrect.

sensitivity, **recall**, **hit rate**, or **true positive rate** (TPR)

$$\mathrm{TPR} = \frac{\mathrm{TP}}{P} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} = 1 - \mathrm{FNR}$$



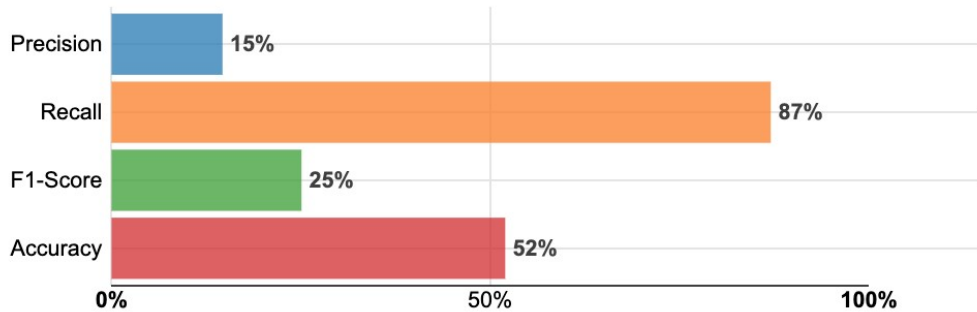
fall-out or **false positive rate** (FPR)

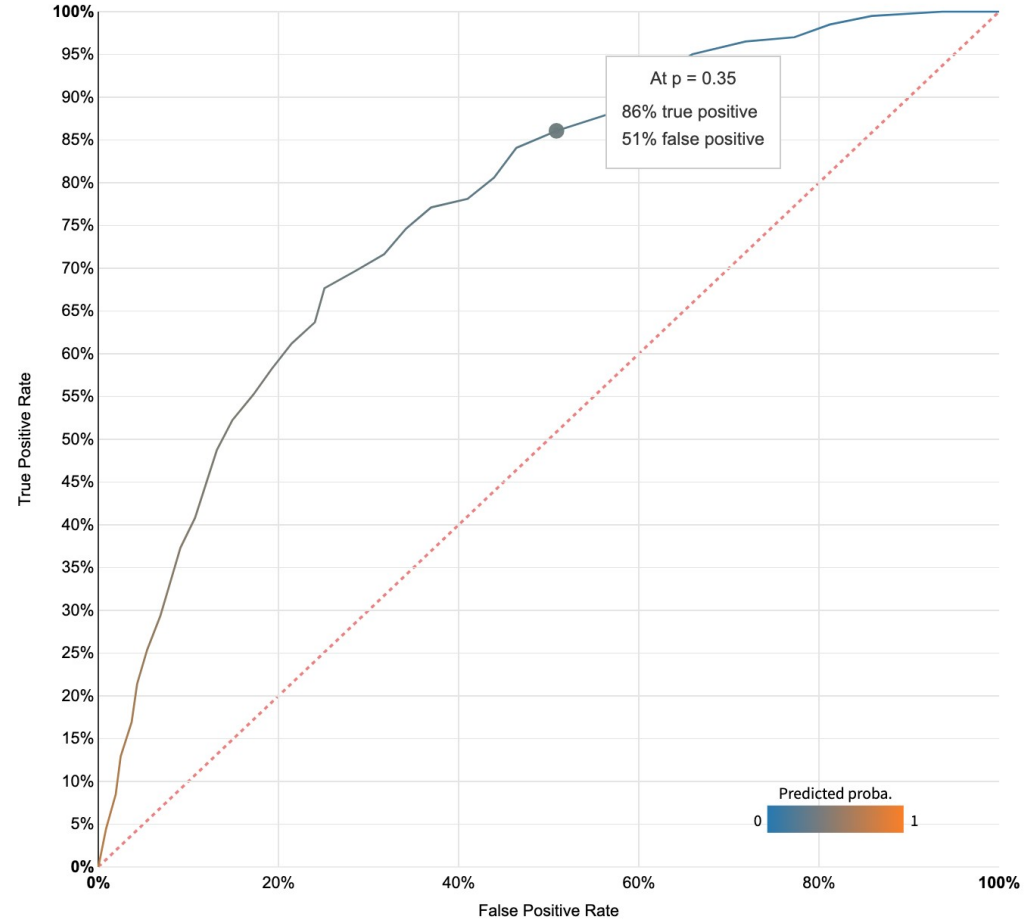
$$\mathrm{FPR} = \frac{\mathrm{FP}}{N} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}} = 1 - \mathrm{TNR}$$

Threshold (cut-off) 0 1 0.350 BACK TO OPTIMAL*

Display: Record count

	Predicted True	Predicted False	Total
Actually True	175	26	201
Actually False	1019	957	1976
Total	1194	983	2177





sensitivity, recall, hit rate, or true positive rate (TPR)

$$\mathrm{TPR} = \frac{\mathrm{TP}}{P} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} = 1 - \mathrm{FNR}$$

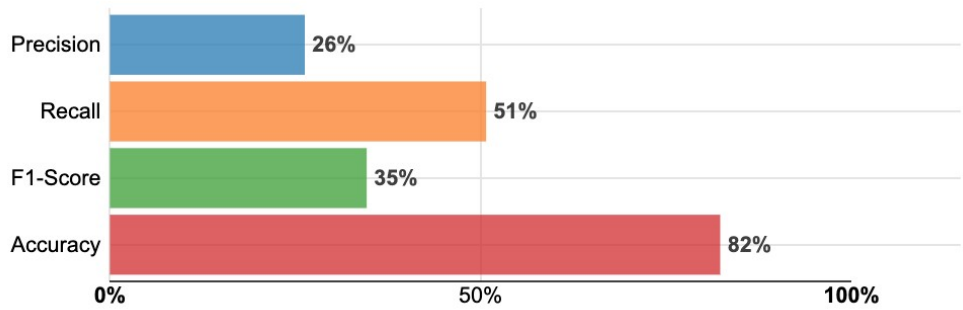
fall-out or false positive rate (FPR)

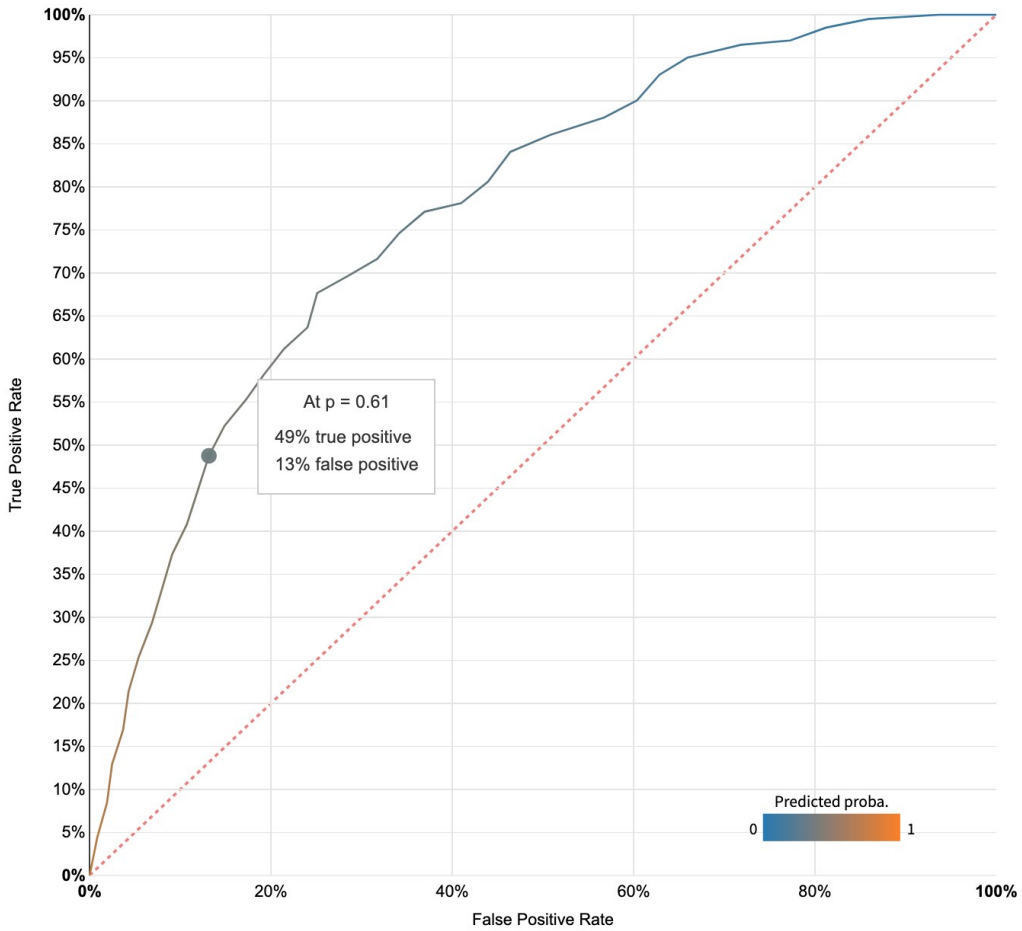
$$\mathrm{FPR} = \frac{\mathrm{FP}}{N} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}} = 1 - \mathrm{TNR}$$

Threshold (cut-off) 0 1 0.600 BACK TO OPTIMAL*

Display: Record count

	Predicted True	Predicted False	Total
Actually True	102	99	201
Actually False	286	1690	1976
Total	388	1789	2177





sensitivity, recall, hit rate, or true positive rate (TPR)

$$\mathrm{TPR} = \frac{\mathrm{TP}}{P} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} = 1 - \mathrm{FNR}$$

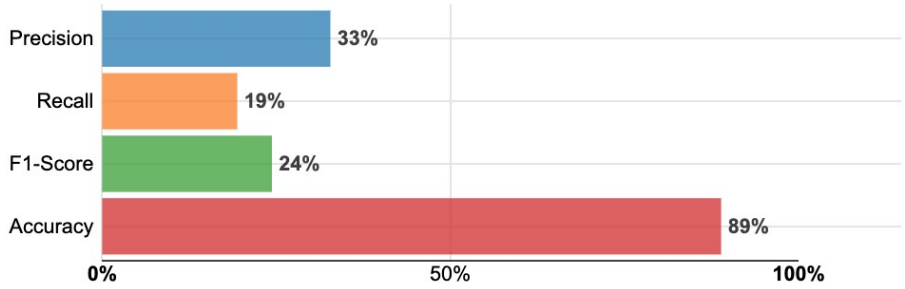
fall-out or false positive rate (FPR)

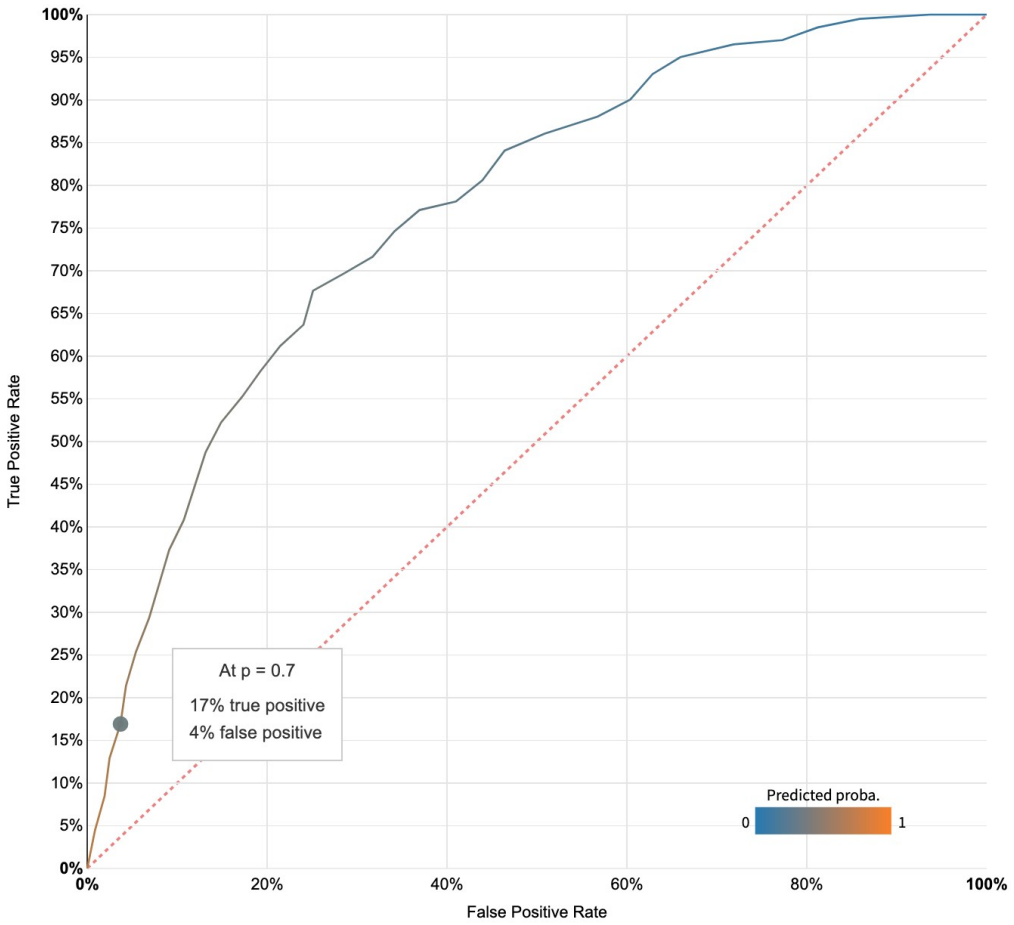
$$\mathrm{FPR} = \frac{\mathrm{FP}}{N} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}} = 1 - \mathrm{TNR}$$

Threshold (cut-off) 0 1 0.700 BACK TO OPTIMAL*

Display: Record count

	Predicted True	Predicted False	Total
Actually True	39	162	201
Actually False	80	1896	1976
Total	119	2058	2177





sensitivity, recall, hit rate, or true positive rate (TPR)

$$\mathrm{TPR} = \frac{\mathrm{TP}}{P} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} = 1 - \mathrm{FNR}$$

fall-out or false positive rate (FPR)

$$\mathrm{FPR} = \frac{\mathrm{FP}}{N} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}} = 1 - \mathrm{TNR}$$